Summary of Current Model Verification Study

Goal

The goal of this round was to find a concrete fermionic model that can support the perturbative SDP improvement framework in the draft. In particular, we wanted to verify three points:

1. exactness of the reduced SDP at a reference Hamiltonian H_0;
2. first-order response matching between the reduced SDP and the physical ground state at H_0;
3. a nontrivial lower bound
$$\|\delta m_S(\theta)\| \geq c_{\mathrm{mis}}|\theta|$$
for selected moments after a weak perturbation $H(\theta) = H_0 + \theta V$.

We tested two spinful Hubbard-type candidates, both based on an isolated Hubbard-dimer reference point.

Common Setup And Notation

- Lattice: an open one-dimensional chain with L sites, L even.
- Filling: half filling, $N_e = L$ electrons.
- Strong dimers: $D_r = (2r-1, 2r)$, $r = 1, \dots, L/2$.
- Weak inter-dimer bonds: $B_r = (2r, 2r+1)$, $r = 1, \dots, L/2-1$.
- Operators:
  - $c_{i,\sigma}, c_{i,\sigma}^\dagger$: fermion annihilation/creation operators at site i with spin $\sigma \in \{\uparrow, \downarrow\}$.
  - $n_{i,\sigma} = c_{i,\sigma}^\dagger c_{i,\sigma}$, and $n_i = n_{i,\uparrow} + n_{i,\downarrow}$.
  - $D_i = n_{i,\uparrow} n_{i,\downarrow}$, the onsite double occupancy.
  - $K_{ij} = \sum_\sigma (c_{i,\sigma}^\dagger c_{j,\sigma} + c_{j,\sigma}^\dagger c_{i,\sigma})$, the bond kinetic observable.
  - $S_i^z = (n_{i,\uparrow} - n_{i,\downarrow})/2$, $S_i^+ = c_{i,\uparrow}^\dagger c_{i,\downarrow}$, and $S_i^- = c_{i,\downarrow}^\dagger c_{i,\uparrow}$.
  - $h.c.$ denotes the Hermitian conjugate.

The common reference Hamiltonian is the isolated-dimer spinful Hubbard model with $t_d = 1$:

$$H_0 = \sum_{r=1}^{L/2} \left[-K_{2r-1,2r} + U(D_{2r-1} + D_{2r}) \right].$$

For response matching, fields are coupled to the selected moments $M_1, \dots, M_s$ by

$$H(\theta, h) = H_0 + \theta V - \sum_{a=1}^{s} h_a M_a.$$

Reported Quantities

The quantities in this subsection are diagnostics used in this numerical report, not standard names from the Hubbard-model literature.

- Reference energy error: $|E_{\mathrm{SDP}}(H_0) - E_{\mathrm{phys}}(H_0)|$.
- Selected-moment error: $\|m_S^{\mathrm{SDP}}(0) - m_S^{\mathrm{phys}}(0)\|$.
- Response-matching error: $\|D_h m_S^{\mathrm{SDP}}(0) - D_h m_S^{\mathrm{phys}}(0)\|$, the norm of the difference between the SDP and physical first-order response matrices.

- Mismatch ratio: $\|\delta m_S(\theta)\|/|\theta|$, where $\delta m_S(\theta) = m_S^{\mathrm{SDP}}(\theta) - m_S^{\mathrm{phys}}(\theta)$. This ratio is used to estimate c_{mis}.
- Solver: all reported SDP values below were computed with MOSEK.

Terminology And References

- The terms "Hubbard model", "spinful Hubbard chain", "hopping", "onsite interaction", "half filling", and "double occupancy" follow the standard Hubbard-model terminology; see Hubbard's original paper and standard treatments of the one-dimensional Hubbard model [1, 2].
- The words "dimerized" and "alternating" refer to chains with alternating strong and weak bonds; this terminology is standard in the dimerized or alternating Hubbard-chain literature [3, 4].
- The term "extended Hubbard model" refers here to adding an intersite density-density interaction to the onsite Hubbard interaction; this is the standard usage in the extended-Hubbard literature [5].
- "Response matrix" is used in the usual static linear-response sense, following the Kubo linear-response terminology [6].
- "Semidefinite programming" follows the standard convex-optimization terminology of Vandenberghe and Boyd [7].

Model 1: Standard Dimerized Spinful Hubbard Chain

Model. The first model is the standard alternating-hopping spinful Hubbard chain. It uses the common reference Hamiltonian H_0 above and the weak inter-dimer hopping perturbation

$$V_{\mathrm{hop}} = -\sum_{r=1}^{L/2-1} K_{2r,2r+1}.$$

The reported scan used $L = 4$, $U = 2$, $t_d = 1$, and $\theta = \pm 0.005, \pm 0.01$.

The selected moment families tested were:

- Bond kinetic moments on strong dimers: $M_r = K_{2r-1,2r}$.
- Double occupancy moments on strong dimers: $M_r = D_{2r-1} + D_{2r}$, and also the site-resolved versions D_{2r-1}, D_{2r}.
- Spin correlation moments on strong dimers: $S^z_{2r-1}S^z_{2r}$, and $\mathbf{S}_{2r-1} \cdot \mathbf{S}_{2r} = S^z_{2r-1}S^z_{2r} + \frac{1}{2}(S^+_{2r-1}S^-_{2r} + S^-_{2r-1}S^+_{2r})$.
- Onsite-pair hopping moments between the two sites of a strong dimer: $P_r = c^\dagger_{2r-1,\uparrow}c^\dagger_{2r-1,\downarrow}c_{2r,\downarrow}c_{2r,\uparrow} + h.c.$.
- Singlet/triplet projectors on a strong dimer: $\Pi_{s,r} = |s_r\rangle\langle s_r|$ and $\Pi_{t0,r} = |t^0_r\rangle\langle t^0_r|$, where $|s_r\rangle = (|\uparrow,\downarrow\rangle - |\downarrow,\uparrow\rangle)/\sqrt{2}$ and $|t^0_r\rangle = (|\uparrow,\downarrow\rangle + |\downarrow,\uparrow\rangle)/\sqrt{2}$ on dimer D_r.
- Bond kinetic moments on weak inter-dimer bonds: $M_r = K_{2r,2r+1}$ on the weak bonds B_r.

Verification results.

1. **Reference exactness.** With MOSEK, the reference energy error was around 10^{-9}.

2. **First-order response matching.** The selected moments supported on strong dimers D_r matched well at H_0. The bond kinetic moments $K_{2r,2r+1}$ on weak inter-dimer bonds gave a large response-matching error.

3. **Mismatch lower bound.** For all observables supported on strong dimers D_r that were tested, including bond kinetic, double occupancy, spin correlations, pair hopping, and singlet/triplet projectors, the mismatch scaled quadratically:

$$\|\delta m_S(\theta)\| \sim C\theta^2,$$

rather than linearly in $|\theta|$. Inter-dimer bond moments did show a strong linear mismatch, while response matching failed for those moments.

Model 2: Density-Perturbed Extended Spinful Hubbard Dimer Chain

Model. The second model uses the same isolated Hubbard-dimer reference H_0, but replaces weak hopping by an inter-dimer density-density perturbation:

$$V_{\text{dens}} = \sum_{r=1}^{L/2-1} n_{2r} n_{2r+1}.$$

The selected moments are the strong-dimer bond kinetic observables:

$$M_r = K_{2r-1,2r}, \qquad r = 1, \dots, L/2.$$

The reported checks used $U = 4$, $t_d = 1$, and $L = 4, 6$.

Verification results.

1. **Reference exactness.** With MOSEK, the reference point was exact to high precision.
   - $L = 4, U = 4$: reference energy error about 5.7×10^{-12}; selected-moment error about 1.0×10^{-7}.
   - $L = 6, U = 4$: reference energy error about 8.2×10^{-13}; selected-moment error about 1.0×10^{-9}.
2. **First-order response matching.** The SDP and physical response matrices had small response-matching errors.
   - $L = 4, U = 4$: response-matching error about 3.0×10^{-5}.
   - $L = 6, U = 4$: response-matching error about 3.5×10^{-6}.
3. **Mismatch lower bound.** The selected bond-kinetic moments showed a stable linear mismatch after the density perturbation.
   - $L = 4, U = 4$: $\|\delta m_S(\theta)\|/|\theta| \approx 0.125$ for $|\theta| \leq 0.02$, corresponding to $c_{\text{mis}} \approx 0.12$ on the tested range.
   - $L = 6, U = 4$: $\|\delta m_S(\theta)\|/|\theta| \approx 0.215$–$0.216$ for $|\theta| = 0.005, 0.01$, corresponding to $c_{\text{mis}} \approx 0.21$ on the tested range.

Observed Status

For the standard hopping perturbation, the reference exactness error was small, the response check for moments supported on strong dimers passed numerically, and the corresponding mismatch was quadratic in θ in the tested cases.

For the density-density perturbation, the tested $L = 4$ and $L = 6$ cases both gave small reference errors, small response-matching errors, and a linear mismatch ratio bounded away from zero on the tested θ-range.

References

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